

BRIDGET KNIGHT

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Technical artist and visualization specialist with a B.S. in Computer Science pursuing an M.A. in Illustration. Skilled in translating complex data into immersive visual storytelling. From developing genomic visualization pipelines to field-based ecological monitoring, I integrate production-level software engineering with a multidisciplinary approach to 3D modeling and visual development. Expert at navigating the full pipeline from raw data and field research to immersive storytelling.

SKILLS

Digital Media: Maya, Blender, Adobe Suite (Ps, Ai, Br, Pr), Clip Studio Paint, Figma, Toon Boom
Engineering: Python, C++, JavaScript, R (Shiny), SQL, HTML/CSS, Linux, Docker, Git/GitHub, Arduino

Technical Art & Research: Iterative design, 3D modeling, scripting, systems documentation, mood board creation and look development
Languages: English (native), French (B1 – Intermediate)

EDUCATION

M.F.A. in Illustration | GPA: 4.0

Savannah College of Art and Design

Jan 2026 - Present

Online

B.S. in Computer Science, Intelligent Systems | GPA: 3.94, Summa Cum Laude

Salem State University

May 2025

Salem, MA

- **Minors:** Mathematics & Physics

- **Relevant Coursework:** Computer Vision, Robotics, Digital Solutions for Illustrators, Game Engine Pipelines (in progress)

PORTFOLIO & PROJECTS

Narrative Environment: Abandoned City Sci-Fi Concept

- Developed 3D blockouts and 10 compositional thumbnails to validate scale; executed final digital painting in Ps/Clip Studio Paint
- Explored lighting and atmosphere keys to strengthen narrative – visible at bridgetknight.com

Physically-Based Black Hole Visualization (Blender & Maya)

- Developed a dynamic gravitational lensing shader in Blender, using node-based logic to simulate Schwarzschild light distortion
- Investigated procedural workflows in Maya using MASH networks to explore cross-platform implementation and constraints
- Configured volumetric rendering and accretion disk texturing for a scientifically grounded aesthetic for astronomical visualization

Sproutling: Automated Plant Watering System

- Built an Arduino-based monitoring system (C++) with integrated sensors and a companion Android app
- Presented a visually compelling scientific poster at the Massachusetts Undergraduate Research Conference (2025)

PROFESSIONAL EXPERIENCE

Field Technician

Mass Audubon / Parker River National Wildlife Refuge

Sep 2025 - Nov 2025

Newburyport, MA

- Conducted field-based environmental monitoring in the Great Marsh to support habitat restoration
- Applied GIS skills to map site pathways and record site data with ArcGIS Field Maps

Software Engineering Intern | Full-time

Broad Institute of MIT and Harvard

Jun 2021 - Jul 2025

Cambridge, MA

- Developed cloud-based search interfaces (Python, HTML, JS) in Terra to analyze TB-scale data for infectious disease researchers
- Led development of automated, reproducible sequencing reports (Python) containerized in Linux for 30+ malaria researchers
- Designed scalable data visualization interfaces using R (Shiny) and Python to improve high-throughput data analysis

Research Assistant

Salem State University – In partnership with MIT

Sep 2023 - Jun 2025

Salem, MA

- Developed Jupyter Notebook pipelines for statistical visualization of physics student data, resulting in a published paper
- Built interactive physics models and labs, translating abstract computational concepts into visual educational tools

HONORS & PUBLICATIONS

- **Scholarships:** SCAD Achievement Honors Scholarship & Student Recognition Scholarship

Sep 2023 - Jun 2025

- **Societies:** Alpha Lambda Delta Honor Society, Upsilon Pi Epsilon (Computer Science)

Joined 2020, 2024

- **Publication:** *Digging In: Attending to students' epistemic emotions while computationally modeling in physics*

Jul 2024

- **Selected Presentations:** *Big Data Visualization and GitHub Insights* at Women in Data Science (2021, 2022); *Automated Sequencing Report Pipelines for Malaria Surveillance and Visualization* at Harvard Medical School (2025)